Cloud Computing: IaaS, PaaS and SaaS

Knowledge document 007 of 100

Introduction to Cloud Computing

Cloud computing is the delivery of computing services, including servers, storage, databases, networking, software and analytics, over the internet on a pay-as-you-go basis. Instead of buying and maintaining their own physical servers, individuals and organisations rent resources from a cloud provider whose huge data centres hold the hardware. The US National Institute of Standards and Technology (NIST) defined cloud computing in 2011 with five essential characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. The word "cloud" comes from the cloud-shaped symbol traditionally used in network diagrams to represent the internet.

History

The idea of sharing computing resources dates back to the 1960s, when John McCarthy suggested that computing might be organised as a public utility. Salesforce delivered business software over the web from 1999. Modern cloud computing took off when Amazon Web Services (AWS) launched Simple Storage Service (S3) in March 2006 and Elastic Compute Cloud (EC2) in August 2006. Google App Engine arrived in 2008, Microsoft Azure launched commercially in 2010, and today AWS, Microsoft Azure and Google Cloud are the three largest providers, with Alibaba Cloud and Oracle Cloud also significant.

The Three Service Models

Cloud services are classified into three main models according to how much the provider manages. IaaS stands for Infrastructure as a Service. It provides basic building blocks: virtual machines, storage and networks. The customer installs and manages the operating system, middleware and applications, while the provider manages the physical hardware. Examples of IaaS include Amazon EC2, Microsoft Azure Virtual Machines and Google Compute Engine. IaaS suits system administrators and companies that need full control.

PaaS stands for Platform as a Service. It provides a ready-to-use platform for building, testing and deploying applications without worrying about servers or operating systems. Developers only write code and manage data. Examples of PaaS include Google App Engine, Heroku, AWS Elastic Beanstalk and Azure App Service. PaaS suits software developers.

SaaS stands for Software as a Service. It delivers complete, ready-to-use applications over the internet, usually through a web browser or app, on subscription. The provider manages everything, and the user simply uses the software. Examples of SaaS include Gmail, Google Docs, Microsoft 365, Zoom, Dropbox and Salesforce. SaaS suits end users. A helpful analogy is pizza: IaaS is

like buying ingredients and cooking at home, PaaS is like ordering a ready base and adding toppings, and SaaS is like eating at a restaurant where everything is done for you. Newer terms include FaaS (Function as a Service), also called serverless computing, where code runs only when triggered, as in AWS Lambda.

Deployment Models

A public cloud is owned by a provider and shared by many customers. A private cloud is dedicated to a single organisation, either on its own premises or hosted. A hybrid cloud combines public and private clouds so data and applications can move between them. Multi-cloud means using services from more than one provider to avoid dependence on one vendor.

Benefits of Cloud Computing

Two key benefits of cloud computing are cost savings and scalability. Cost: customers avoid large upfront investments in hardware and pay only for what they use, turning capital expenditure into operating expenditure. Scalability and elasticity: resources can be increased within minutes during heavy demand, such as a festival sale, and reduced afterwards. Other benefits include accessibility from anywhere with internet, automatic backups and disaster recovery, faster deployment of new projects, high reliability through multiple data centres, and automatic software updates. Cloud platforms also provide tools for artificial intelligence, such as GPUs for training models, which students and researchers can access through services like Google Colab.

Cloud Security: The Shared Responsibility Model

Security in the cloud is shared. The provider secures the physical data centres, hardware and core services, while the customer is responsible for securing their own data, user accounts, passwords and configuration. Many cloud data breaches happen because of misconfigured storage or weak access control rather than failures of the provider. Best practices include strong passwords with multi-factor authentication, encryption, least-privilege access and regular backups.

Uses and Indian Examples

Cloud computing powers video streaming, online banking, e-commerce, education platforms and AI services. In India, the government's MeghRaj cloud initiative, launched in 2014, promotes cloud use in government departments. Global providers have built data-centre regions in India: AWS opened a Mumbai region in 2016 and a Hyderabad region in 2022, while Microsoft and Google also run Indian regions. Chennai has become an important data-centre hub because several submarine internet cables land there. Startups such as Zoho, based in Chennai, offer SaaS products including Zoho CRM and Zoho Mail to customers worldwide, and UPI and e-commerce platforms rely heavily on cloud infrastructure.

Challenges and Future

Challenges include data privacy and data residency laws, internet dependence, vendor lock-in, unexpected costs and security misconfigurations. Future trends include edge computing, which processes data closer to devices, serverless architectures, AI-optimised cloud infrastructure and sustainable green data centres.

Quick Facts

- SaaS stands for Software as a Service; IaaS is Infrastructure as a Service and PaaS is Platform as a Service.
- Examples: Amazon EC2 for IaaS, Heroku for PaaS and Gmail for SaaS.
- Two benefits of cloud computing are lower upfront cost and easy scalability.
- AWS launched S3 and EC2 in 2006, beginning the modern cloud era.
- The cloud security model is shared: the provider secures the infrastructure and the customer secures data and access.